B.Sc. (HONS.) IN CSE FIRST YEAR, SECOND SEMESTER EXAMINATION, 2021

DIGITAL SYSTEMS DESIGN

[According to the New Syllabus]

Subject Code : 510221

Examination Code : 5612

Time—3 hours

Full marks—80

[N.B. The figures in the right margin indicate full marks. Answer any **four** *questions. Different part of a questions must be answered sequentially.]*

Marks

1. (*a*) What is analog and digital system? Write down the advantages of digital techniques. 5

   (*b*) Convert the following numbers into given base numbers : 6

   (*i*) $(110101)_2 = (\,?\,)_{10}$

   (*ii*) $(129A{\cdot}B8C)_{16} = (\,?\,)_2$

   (*iii*) $(225{\cdot}225)_{10} = (\,?\,)_2$.

   (*c*) Explain the method of subtraction using 2′s complement method with example. 5

   (*d*) Write short notes about the following terms : 4

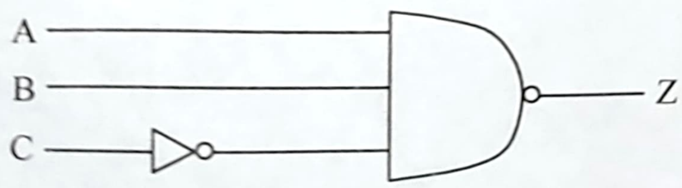
   (*i*) ASCII Code

   (*ii*) Gray Code.

2. (*a*) Define universal gate. Prove the universality of NAND gate and NOR gate. 6

   (*b*) State and prove De-Morgan's theorem. 5

   (*c*) Determine the output expression for the circuit of the following figure and simplify it using De-Morgan's theorem. 4



   (*d*) Simplify the expression. 5

$$Z = ABC + A\bar{B} \cdot \left(\overline{\bar{A} \cdot \bar{C}}\right).$$

[Please turn over

		Marks
3.	(*a*) What is Flip-Flop? Write down the applications of Flip-Flop.	3
	(*b*) Explain a clocked J-K Flip-Flop.	6
	(*c*) Define NOR gate latch? Explain clocked S-C Flip-Flop.	6
	(*d*) Explain how J-K Flip-Flop can be converted to : (*i*) D-type Flip-Flop (*ii*) T-type Flip-Flop.	5
4.	(*a*) To construct a Full Adder circuit by using Half-Adder circuit.	6
	(*b*) What is Encoder? Design a Octal to Binary Encoder?	6
	(*c*) What is ripple counter? Explain a 4-bit ripple counter with diagram.	5
	(*d*) Write down the difference between synchronous and asynchronous.	3
5.	(*a*) What is Demultiplexer? Sketch the logic diagram with truth table of 1 to 8 line Demultiplexer.	6
	(*b*) What is NAND latch? Describe S-R Flip-Flop with logic diagram.	6
	(*c*) Draw and explain 5-bit shift register.	5
	(*d*) Mention the major drawbacks of S-R Flip-Flop.	3
6.	(*a*) Outline the steps that take place when the CPU reads from memory.	5
	(*b*) Differentiate between EPROM and EEPROM.	5
	(*c*) Describe the basic operation of the digital ramp ADC.	6
	(*d*) What is the major advantages of SAC over a digital ramp ADC?	4